

Overview

This tool evaluates fairness in machine learning models by detecting individual discriminatory instances (IDIs). It generates paired inputs that differ only in a sensitive feature (age) and checks if the model produces different outputs.

Files

- fairness_tool.py: Runs the fairness testing experiment
- plot_graph.py: Generates a graph from the results
- fairness_results.csv: Raw experimental results
- fairness_summary.csv: Aggregated results
- idi_plot.png: Graph of IDI ratio vs budget

How to Run

1. Install dependencies:
`pip install -r requirements.txt`
2. Run the fairness testing:
`python fairness_tool.py`
3. Generate the graph:
`python plot_graph.py`

Output

- fairness_results.csv
- fairness_summary.csv
- idi_plot.png

These outputs are used for analysis in the report.